C++ LEVEL 2

Revision Question Bank

50 practice problems • Intermediate • C++17

Topics covered
Arrays, strings, functions, pointers, memory management, recursion, searching, sorting, matrices, and STL.
How to use this workbook
Attempt each question before looking up a solution. Compile with warnings enabled, test edge cases, and explain your approach and complexity where relevant.
Compiler suggestion
`g++ -std=c++17 -Wall -Wextra -Wpedantic file.cpp -o program`

Name: _____ Date started: _____

Module 1 — Arrays

1. Find the Largest Element

Given an array of integers, find the largest element without sorting the array.

2. Find the Smallest Element

Find the minimum element in an array.

3. Calculate Array Sum and Average

Calculate the sum and average of all elements.

4. Reverse an Array

Reverse an array in place without creating another array.

5. Find the Second-Largest Element

Find the second-largest distinct element without sorting.

6. Count Even and Odd Elements

Count the even and odd values in an array.

7. Linear Search

Find a target element and return its index, or indicate that it is absent.

8. Remove Duplicates from a Sorted Array

Modify a sorted array in place so each distinct value appears once.

9. Rotate an Array

Rotate an array to the right by k positions.

10. Move All Zeros to the End

Move zeros to the end while preserving the relative order of nonzero elements.

Module 2 — Strings

11. Reverse a String

Reverse a string without using a built-in reverse function.

12. Check a String Palindrome

Determine whether a string reads the same forward and backward.

13. Count Vowels and Consonants

Count vowels and consonants in a string, handling uppercase and lowercase letters.

14. Count Words in a Sentence

Count words while handling repeated spaces and leading or trailing whitespace.

15. Character Frequency

Count how many times each character occurs in a string.

16. Check Anagrams

Determine whether two strings contain the same characters with the same frequencies.

17. Remove Duplicate Characters

Remove repeated characters while preserving the first occurrence of each character.

18. First Non-Repeating Character

Find the first character that appears exactly once.

Module 3 — Functions and Modular Programming

19. Build a Calculator Using Functions

Implement addition, subtraction, multiplication, and division in separate functions.

20. GCD and LCM Functions

Implement separate functions to calculate the GCD and LCM of two integers.

21. Prime Numbers Using Functions

Create a function that checks whether a number is prime and use it to print primes in a range.

22. Pass by Value vs. Pass by Reference

Write a program that demonstrates the difference between passing arguments by value and by reference.

23. Swap Using References

Write a function that swaps two integers using reference parameters.

24. Function Overloading

Create overloaded functions that calculate the area of a circle, rectangle, and square.

Module 4 — Pointers and Memory

25. Pointer Basics

Declare an integer and a pointer to it. Print the value, address, and dereferenced value.

26. Modify a Variable Through a Pointer

Write a function that changes a variable's value using a pointer parameter.

27. Traverse an Array Using Pointers

Print all array elements using pointer arithmetic instead of array indexing.

28. Find the Largest Element Using Pointers

Find the maximum array element using pointers.

29. Dynamic Memory Allocation

Allocate an array dynamically using `new[]`, initialize it, print its values, and release the memory correctly.

30. Pointer Safety

Demonstrate and explain null pointers, dangling pointers, and why dereferencing invalid pointers is unsafe. Write a safe example using `nullptr`.

Module 5 — Recursion

31. Factorial Using Recursion

Calculate the factorial of a non-negative integer recursively.

32. Fibonacci Using Recursion

Generate Fibonacci numbers using a recursive function.

33. Sum of Digits Using Recursion

Calculate the sum of an integer's digits recursively.

34. Reverse a String Recursively

Reverse a string using recursion without a built-in reverse function.

35. Calculate Power Recursively

Calculate a^b recursively for a non-negative exponent.

36. Binary Search Recursively

Implement binary search using recursion on a sorted array.

37. Tower of Hanoi

Print the steps required to move n disks between three rods, following the puzzle rules.

Module 6 — Searching and Sorting

38. Binary Search Iteratively

Find a target in a sorted array using binary search.

39. Bubble Sort

Sort an array in ascending order using bubble sort.

40. Selection Sort

Sort an array using selection sort.

41. Insertion Sort

Implement insertion sort and test it on sorted, reversed, and randomly ordered input.

42. Merge Two Sorted Arrays

Merge two sorted arrays into a single sorted array.

43. Merge Sort

Implement merge sort using divide and conquer.

44. Quick Sort

Implement quicksort and explain how pivot selection affects performance.

Module 7 — Matrices, STL, and Problem-Solving

45. Matrix Addition and Transposition

Add two compatible matrices and calculate the transpose of a matrix.

46. Matrix Multiplication

Multiply two matrices when their dimensions are compatible.

47. Frequency Counter Using std::unordered_map

Count the frequency of each integer in an array using the C++ Standard Library.

48. Remove Duplicates Using std::set

Remove duplicate values from a collection and print the unique values in sorted order.

49. Balanced Parentheses

Given a string containing (), {}, and [], determine whether the brackets are correctly balanced. Use `std::stack`.

50. Two Sum

Given an integer array and a target, find two distinct elements whose sum equals the target. Try a brute-force solution first, then improve it using `std::unordered_map`.

Completion checklist

Mark a question complete when your solution passes your own tests. Aim to explain your approach and test edge cases before moving on.

■ Q1	■ Q2	■ Q3	■ Q4	■ Q5
■ Q6	■ Q7	■ Q8	■ Q9	■ Q10
■ Q11	■ Q12	■ Q13	■ Q14	■ Q15
■ Q16	■ Q17	■ Q18	■ Q19	■ Q20
■ Q21	■ Q22	■ Q23	■ Q24	■ Q25
■ Q26	■ Q27	■ Q28	■ Q29	■ Q30
■ Q31	■ Q32	■ Q33	■ Q34	■ Q35
■ Q36	■ Q37	■ Q38	■ Q39	■ Q40
■ Q41	■ Q42	■ Q43	■ Q44	■ Q45
■ Q46	■ Q47	■ Q48	■ Q49	■ Q50